

## Template for Computer Programming I Practical Recording

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| <b>Date</b>                  | : | 30/01/2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Problem Specification</b> | : | Write a java program for the game Rock, Paper, Scissors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Assumption</b>            | : | The user inputs are valid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Limitation</b>            | : | random integers are between 0 and 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Input</b>                 | : | The user enters their play, while the computer selects a random play.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Processing</b>            | : | The system compares the choices and declares the winner.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Output</b>                | : | USER win or loss                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Algorithm</b>             | : | <p>Step1: Define the variables as computerinput and computerplay.</p> <p>Step2: Generate a random value( <math>0 \leq \text{computerinput} \leq 2</math> ) for computerInput using a random function.</p> <p>Step3: convert the computerinput into computerpaly using switch case</p> <div style="margin-left: 40px;"><p>0 ← R</p><p>1 ← P</p><p>2 ← S</p></div> <p>Step4: prompt "Enter your play R or P or S "</p> <p>Step5: Define the variables as userplay</p> <p>Step6: Read the userplay given by user</p> <p>Step7: Assign input value for userplay</p> <p>Step8: print computerplay as "computerplay is : "</p> <p>Step9: if(computerplay.equals("S") &amp;&amp; userplay.equals("R")) is True<br/>Print " Rock crushes scissors, you win!".then go to step16</p> <p>Step10: else if(computerplay.equals("P") &amp;&amp; userplay.equals("R")) Is True<br/>print " you loss!" .then go to step16</p> <p>Step11: else if(computerplay.equals("S") &amp;&amp; userplay.equals("P")) is True<br/>print " scissors cuts the paper ,you loss!" .then go to step16</p> <p>Step12:else if(computerplay.equals("R") &amp;&amp; userplay.equals("P")) is True<br/>print "you win!" .then go to step16</p> <p>Step13:else if(computerplay.equals("R") &amp;&amp; userplay.equals("S")) is True<br/>print " Rock crushes scissors,you loss!" .then go to step16</p> |

Step14:else if(computerplay.equals("P") && userplay.equals("S")) is  
True  
print " scissors cuts the paper ,you win!" .then go to step16

Step15:else  
Print "Both are same.Try again"

Step16: End the Programme

|                                      |   |                                                                                                                                                    |
|--------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programme listing</b>             | : | Programme file <b>attached</b>                                                                                                                     |
| <b>Test data and expected output</b> | : | 1. Test data:           userplay= "R"<br>Computerinput= P<br>Expected output:   loss!                                                              |
| <b>Output obtained for test data</b> | : | 1. Test data:           userplay= "R"<br>Computerinput= P<br>Obtained output:   loss!                                                              |
| <b>Discussion</b>                    | : | It compares the user's and computer's choices and announces the win and loss based on the rules of the game.                                       |
| <b>Conclusion</b>                    | : | The user inputs their choice and the computer randomly picks choice. It compares the two and prints the result (win/loss) based on the game rules. |